

Diurnal and Daily Variations of PM_{2.5} and its Multiple-Wavelet Coherence with Meteorological Variables in Indonesia

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ABSTRACT

PM_{2.5} is a fine particle that has adverse health effects. Characterizing the diurnal variations and the influence of meteorology is critical for understanding the drivers of air pollution and planning effective mitigation strategies. We studied the diurnal variation of PM_{2.5} and its relationship with meteorological variables in seven cities representing Indonesia's three different rainfall patterns during 2021. We used half-hourly PM_{2.5} concentrations obtained by air quality monitoring system (AQMS), planetary boundary layer height (PBLH) estimation from radiosonde, and meteorological parameters from meteorological stations. A bimodal pattern with two peaks appears in Padang, Manado, Palu, and Pangkalpinang, while in Jakarta, Surabaya, and Pontianak, the diurnals have a unimodal pattern with one peak at night until morning. All cities generally present higher diurnal PM_{2.5} concentrations in the dry season than in the wet season. The relationship between PM_{2.5} concentration and PBLH shows Jakarta, Surabaya, Padang, and Pontianak have a strong anti-correlation for different seasons, while the unusual positive correlation occurs in Padang. The Pearson correlation between PM_{2.5} concentration with each meteorological factor is significant in monthly data and insignificant in daily data. Implementation of Multiple Wavelet Coherence (MWC) with various meteorological variables reveals that the combination of four parameters provides a stronger influence on the PM_{2.5} concentration in all the observed locations. Wavelet analysis also observes distinct scale periods that represent higher haze concentrations in Jakarta and Surabaya from May to September. Meanwhile, the investigation on the extreme rise of PM_{2.5} in Pontianak due to peatland forest fires using HYSPLIT shows that emission from the surrounding area significantly raises the maximum half-hourly in Pontianak to 700 µg m⁻³.

Keywords: PM_{2.5}, Diurnal variation, Meteorological factor, Multiple-wavelet coherence, Indonesia

1 INTRODUCTION

Airborne particulate matter (PM) with a diameter of 2.5 micrometers or less (PM_{2.5}) contributes more to adverse health effects as it can penetrate the lung and spread to the whole body (Murray *et al.*, 2020; Chowdhury *et al.*, 2022). Characterizing the diurnal variations and the influence of meteorology factors is critical for understanding the drivers of air pollution and planning effective mitigation strategies (Schnell *et al.*, 2018). Preceding studies on estimated health risks of exposure

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to PM_{2.5} in Indonesia utilized model analysis or satellite data and primarily focused on long-term exposure (e.g., [Suryadhi et al., 2020](#); [Yu et al., 2020](#)). Long-term in-situ observation studies are essential for an improved understanding of these factors. Several studies presented long-term observations of PM_{2.5} in Indonesia's urban areas (e.g., [Cholianawati, 2022](#); [Santoso et al., 2020](#)), but the diurnal characteristics and the relation with meteorology factors remained unexplored. Diurnal variation in PM_{2.5} is highly uncertain due to many factors which is normally difficult to be well resembled by chemical transport modeling studies ([Chen et al., 2020b](#)). This information is scarce in Indonesia, and models typically use a global standard profile to represent conditions in Indonesia (e.g., [Kombara and Komala, 2023](#)). Studies on the time scale correlation between PM_{2.5} and its influencing factors, such as air pollutants and meteorological factors, are necessary to enhance model performance significantly ([Wang et al., 2020](#)). Individual meteorological factors correlate with PM_{2.5} only at broader time scales. Therefore, MWC is required to analyze multiple meteorological factors affecting the PM_{2.5} concentrations at the same time ([Zhang et al., 2022b](#)).

According to hourly measurements and depending on the monitoring site background, the global diurnal PM_{2.5} cycles show typical either bimodal or unimodal patterns. The bimodal pattern is characterized by morning and night peaks with an afternoon low point ([Manning et al., 2018](#)). Meanwhile, both patterns were present simultaneously in Beijing. PM_{2.5} was recorded to show a bimodal pattern in Beijing's urban areas, while a unimodal pattern was observed in rural areas ([Zhao et al., 2009](#)). Minimum early afternoon concentrations indicate good dispersion with a high daytime planetary boundary layer height (PBLH) accompanied by strong wind. At the same time, the morning peak is attributed to enhanced anthropogenic activity during rush hours with a relatively shallow mixing height. The enhanced emission sources, the lowered PBLH, and wind speed in the late afternoon yielded high concentrations during evening hours. The diurnal variability of the boundary layer, rainfall variability, and emission source strength strongly influence the diurnal variations of PM_{2.5} in the urban area. The diurnal wind patterns and long-range emissions transport mostly dominate the PM_{2.5} variation in the rural area.

Relevant studies investigating the influence between PM_{2.5} variations and other factors generally use time-frequency analysis. Time series analysis using Pearson linear correlation is used to analyze trend analysis and measure the linear relationship between PM_{2.5} and other factors that want to be investigated, such as the source of emissions from the increase in motorized vehicles ([de Jesus et al., 2020](#)). PM_{2.5} concentration is a complex non-stationary signal, so the logic is that not just one factor, but various factors will always impact it simultaneously ([Li et al., 2021](#)). Moreover, the existence of local PM_{2.5} oscillation with a high frequency of occurrence in short periods (as well as the opposite situation) in time series data causes the use of wavelets often conducted to analyze the effects of local variations in PM_{2.5} about other factors on a long-range dataset ([Wu et al., 2023](#)). For example, by applying Continuous Wavelet Transform (CWT) for three years observation dataset in Chengdu City, China, multiple oscillation periods in the real part of wavelet coefficient indicate haze on the scale of 14–32 d, 62–104 d, 105–178 d, and 216–389 d ([Chen et al., 2020c](#)). Extended wavelet in the form of multiple wavelet coherence is used to provide quantitative analysis on the relation between PM_{2.5} variation with a combination of various factors. The extended wavelet coherence between PM_{2.5} with single factors (Wavelet Transform Coherence) and multiple factors (Multiple Wavelet Coherence) are commonly used methods. Multiple meteorological factors affect the PM_{2.5} concentrations at the same time. By applying MWC for investigating the relationship between PM_{2.5} concentrations and meteorology in Xi'an (China), it was concluded that the coherence values for multiple meteorological factors are significantly higher than those for a single meteorological factor, indicating that the variation of PM_{2.5} concentrations is determined by multiple variables ([Zhang et al., 2022b](#)).

A study in Indonesia showed that the mean annual PM_{2.5} concentrations observed in Java's important cities exceeded the annual ambient air quality standard (15 µg m⁻³) ([Santoso et al., 2020](#)). The total emission burden for primary PM_{2.5} in Jakarta is around 4.6 kton, generally emitted from on-road vehicles and industrial combustion sectors ([Lestari et al., 2020](#)). The frequent fires in degraded forests and peatlands have also contributed to substantial increases in PM_{2.5} and influenced its diurnal pattern ([Siregar et al., 2022](#); [Santoso et al., 2022](#)). According to a report by [Dihni \(2022\)](#), Indonesia's forest and land fires in 2021 covered an area of 354,582 ha. However, Jakarta was among the unaffected provinces. Large cities in Sumatra and Kalimantan Islands are



commonly highly affected by forest fires (mainly from peatland burning), e.g., Jambi, Riau, Palangkaraya, and Pontianak. Therefore, accurate time variation information on $PM_{2.5}$ concentrations is highly required.

To the best of our knowledge, this study pioneers deep analysis of the diurnal variations of $PM_{2.5}$ and its relationship with meteorological variables in Indonesia. This was made possible by collecting half-hourly in-situ data from AQMS operated at seven cities in the western and eastern parts of Indonesia for 2021. Note that the year selection may be affected by the large-scale social restrictions due to the COVID-19 pandemic. However, the dataset generated for this year is the best available and complete database of $PM_{2.5}$ concentration and other meteorological parameters. The cities in the analysis also represent three different rainfall patterns in the country: monsoonal, equatorial, and local. The diurnal patterns were analyzed to investigate whether bimodal patterns in the time series data occurred consistently in all study locations. The influence of PBLH on each city's diurnal variation is examined to investigate the dominant factor contributing to the pattern. A continuous wavelet transform (CWT) is used to reveal the oscillation period of haze pollution due to $PM_{2.5}$, which is beneficial for prediction purposes. The backward trajectory was also conducted using the Hybrid Single-Particle Lagrangian Integrated Trajectory Model (HYSPPLIT) to analyze potential sources, transmission paths, and emission contribution during extreme $PM_{2.5}$ concentrations.

2 METHODS

2.1 Data and Site Description

This study uses time-series data of half-hourly $PM_{2.5}$ concentrations from January to December 2021 obtained from AQMS stations managed by the Indonesian Ministry of Environment and Forestry (MoEF). These stations monitor $PM_{2.5}$ and air pollutant gases such as NO_x , SO_2 , and CO with periodic quality assurance and quality control (QA/QC) efforts. The data have been used to generate the official public air quality index (AQI) for the country (<https://ppkl.menlhk.go.id/website/index.php?iku=on>) and utilized in several publications (Rendana, 2021; Verma *et al.*, 2023).

The location and year of the study are selected based on the availability of a minimum of 200 out of 365 days without consecutive periods of unavailability exceeding 20 days. We considered this to be the best available dataset generated by the stations. The radiosonde data, consisting of the vertical distribution of pressure, temperature, humidity, wind speed, and direction at 00:00 and 12:00 UTC per day, were obtained from the Wyoming University website (www.weather.uwyo.edu/upperair). Surface meteorology data (wind speed, rainfall, relative humidity, sun duration, and temperature) from an automatic weather station (AWS) were taken from the Indonesian National Meteorological, Climatological, and Geophysical Agency (<https://dataonline.bmkg.go.id>). Table 1 highlights the coordinates and distance between the AQMS and AWS. Supporting data such as total area and population were obtained from reports of the Central Agency on Statistics.

Jakarta and Surabaya are urban metropolitan areas, while others are large and medium-scale urban areas. Padang, Pangkalpinang, Jakarta, Surabaya, and Pontianak are in the western part of Indonesia, while Palu and Manado reside in the eastern part, as illustrated in Fig. 1. The location of the sites is also linked to Indonesia's climate rainfall patterns categorized into monsoonal, equatorial, and local patterns. Each of them has significant differences in meteorology, which

Table 1. The coordinate location of AQMS, AWS, and distance between them.

City	AQMS	AWS	Distance (km)
Jakarta	(-6.21, 106.8)	(-6.15, 106.8)	7.7
Manado	(1.48, 124.8)	(1.44, 124.8)	4.6
Padang	(-0.93, 100.4)	(-0.99, 100.4)	6.4
Palu	(-0.88, 119.9)	(-0.915, 119.9)	5.3
Pangkalpinang	(-2.15, 106.1)	(-2.17, 106.1)	3.1
Pontianak	(-0.079, 109.3)	(-0.03, 109.3)	5.6
Surabaya	(-7.26, 112.7)	(-7.20, 112.7)	8.7

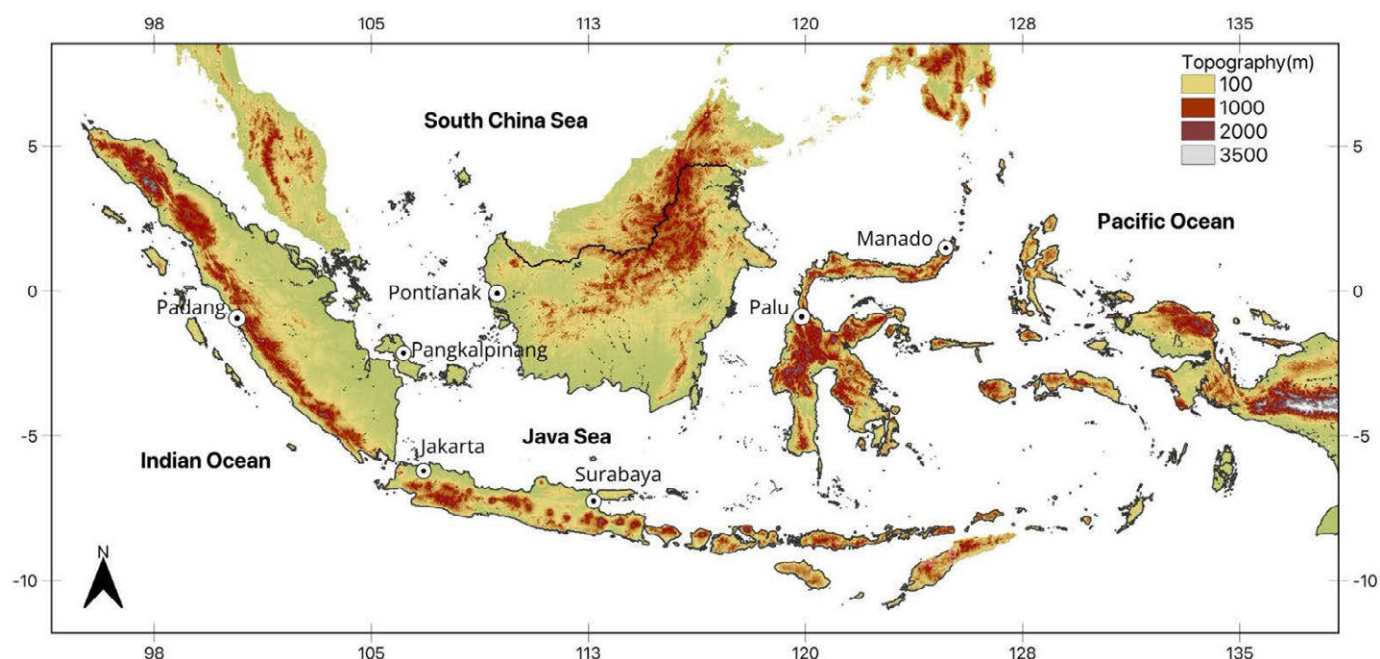


Fig. 1. The seven PM_{2.5} monitoring sites in Indonesia used in this study.

affects the atmospheric compositions, including PM_{2.5} concentration. The study locations represent all climate categories. The monsoonal rainfall patterns with one peak during December–January–February (DJF) occur in Jakarta, Surabaya, and Manado. On the other hand, an equatorial rainfall pattern with two peaks in March–April–May (MAM) and September–October–November (SON) takes place in Padang, Pontianak, and Pangkalpinang. Meanwhile, among the others, only Palu poses an anti-monsoonal (local) pattern with one peak in June–July–August (JJA) (Faidah *et al.*, 2022; Pujiastuti and Nurjani, 2018). When considering topographic factors, the meteorological condition in Padang is the most influenced by topography since it is located near the Barisan Mountains where the land-sea breeze is the major driver. Thus, the rainfall has no significant variation along the year (As-syakur *et al.*, 2016; Marzuki *et al.*, 2021).

2.2 Method to Determine PBLH

PBLH imposes a significant influence on ground-level PM_{2.5} concentration (Tursumbayeva *et al.*, 2022). Several methods have previously been utilized, such as Bulk Richardson number (*Ri*) and the 1.5-theta-increase (*PT*) methods, which were proven more practical for sounding data (Yang *et al.*, 2019). The *PT* method defines the *z_i* as the height where the potential temperature first exceeds the minimum potential temperature within the boundary layer by 1.5 K (Hu *et al.*, 2019). *Ri* is a dimensionless number associating vertical stability with vertical shear and describes the ratio of thermally produced turbulence to that generated by vertical shear. This method assumes no turbulence generation at the top of the stable boundary layer (SBL). Since wind-shear-generated turbulence is significantly reduced above the top of the atmospheric boundary layer, *Ri* increases significantly at the top of SBL (Seibert, 2000). *Ri* at height *z* above ground level can be calculated from sounding data using Eq. (1) (Zhang *et al.*, 2022a).

$$Ri = \left(\frac{gz}{\theta_{v0}} \right) \left(\frac{\theta_{vz} - \theta_{v0}}{u_z^2 + v_z^2} \right) \quad (1)$$

where *g* is gravity acceleration; θ_{v0} and θ_{vz} are the virtual potential temperature at the surface and height *z*, respectively; and *u_z* and *v_z* are the wind speed components at height *z*. The relationship between PBLH and PM_{2.5} concentrations is computed using Pearson's linear correlation (*R*) coefficients.



2.3 Method to Analyze the Influence of Meteorological Factors

The influence of meteorological factors on PM_{2.5} concentration is well-known (Chen *et al.*, 2020c). In this study, Pearson correlation and MWC were performed to investigate the relationship between PM_{2.5} characteristics and meteorological factors in Indonesia. Pearson correlation has often been used to measure the correlation between the two-time series of PM_{2.5} and a single parameter of meteorological factor (Hien *et al.*, 2019).

MWC is applied in this study to analyze the effect of multiple factors on the PM_{2.5} concentration build-up, which, in this case, shows the effect of the combination of meteorological factors. Previous studies have investigated whether the MWC could notice which combination of meteorological factors influences the temporal changes of PM_{2.5}, as explained by Eq. (2) (Li *et al.*, 2021; Zhang *et al.*, 2022b).

$$\rho_m^2(s, \tau) = \frac{w^{Y,X}(s, \tau) \leftrightarrow w^{X,X}(s, \tau) \leftrightarrow w^{Y,X}(s, \tau)^*}{\leftrightarrow w^{Y,Y}(s, \tau)} \quad (2)$$

where $w^{Y,X}(s, \tau)$ is the smoothed cross wavelet power spectra with Y as dependent variable (PM_{2.5}), and X as the independent variable (meteorological factors). $w^{Y,X}(s, \tau)^*$ is the complex conjugate of $w^{Y,X}(s, \tau)$. s and τ represent the scale and time domain location, respectively. In MWC, the cone of influence (COI) separates the wavelet power spectrum with the distorted value due to the edge effect of the wavelet (Ng and Chan, 2012; Nugroho *et al.*, 2021).

To statistically analyze the result of MWC, an investigation on the average wavelet coherence (AWC) and the percent area of significant coherence (PASC) is performed. In MWC, utilization of several predictor variables is feasible; the larger number of variables will increase the width of the power spectrum but not necessarily with the coherence level. AWC is related to the average of the power spectrum, while the PASC confirms the percentage of coherence level. PASC is the division of the average power spectrum within the coherence level area by the average of all the power spectrums within the MWC. The more predictor variable will increase AWC, but not necessarily with the PASC. Only the number of significant predictors will raise the PASC (Hu and Si, 2016). Higher AWC and PASC indicate a greater variation of PM_{2.5} in relation to a particular combination of meteorological factors.

2.4 Methods for Haze Oscillation Period Analysis

The oscillation periods and the evolutionary features of PM_{2.5} are also essential for haze study and prediction (Chen *et al.*, 2020a). Therefore, the wavelet method was applied to reveal the oscillation period. In the wavelet application, several factors were considered, such as data type and target of analysis. Each time series has a random periodic oscillation signal with different frequencies at various scales. A non-orthogonal and complex wavelet function is the most suitable to capture the localized oscillation, amplitude, and phase of the PM_{2.5} time series data (Meng and Sun, 2021). On the other hand, a wavelet function yields only one component, which makes it helpful in isolating discontinuities or peaks of PM_{2.5} time series (Chen *et al.*, 2020c).

Based on these considerations, a non-orthogonal continuous wavelet transform (CWT) with a Morlet wavelet function was used in this study. The real part of the CWT is useful to observe the distribution of the wavelet power spectrum (maximum, minimum, and transition) of PM_{2.5} over scale, time, and frequency. The wavelet power spectrum obtained from the wavelet transform's absolute squared value expresses the correlation between the wavelet function and the time series data.

2.5 Method to Analyze Emission Source Trajectory

The pollutant sources within and around the city, particularly during the dry season, will impact its diurnal concentration of PM_{2.5}. In this work, we conducted a backward trajectory analysis that quantitatively estimates the air mass direction linked to PM_{2.5} sources using the HYSPLIT. This tool has been widely used to forecast and hindcast pollutant transport and dispersion (Bhatti *et al.*, 2021; Li *et al.*, 2020), the potential source of long-range pollutant distribution and transmission paths



(Li *et al.*, 2020; Cui *et al.*, 2021). HYSPLIT is used in this study to investigate the sources and their contributions to the PM_{2.5} diurnal cycle anomaly in Pontianak as a biomass-burning region recorded by the observation data. Meteorological data used for running the HYSPLIT model were gathered from the global data assimilation system (GDAS) from the National Center for Environmental Prediction (NCEP) with a resolution of $1^\circ \times 1^\circ$.

3 RESULTS

3.1 Diurnal Pattern and the Influence of PBLH

The profiles of each site, including their annual average and diurnal variability of PM_{2.5} concentrations in 2021, are detailed in Table 2. The highest PM_{2.5} annual concentrations were observed in the metropolitan cities of Jakarta and Surabaya, showing the impacts of primary anthropogenic PM_{2.5} emission mainly from fossil fuel combustion sources. The annual average level in Surabaya is consistent from 2020 to 2021, while in Jakarta, a rise is noted from $27.84 \mu\text{g m}^{-3}$ in 2020 to $33.9 \mu\text{g m}^{-3}$ in 2021, which were well above the national ambient air quality standards (NAAQs) of $15 \mu\text{g m}^{-3}$. A high level of PM_{2.5}, which was affected by the forest fire during the dry season, was also observed in Pontianak. Meanwhile, levels of PM_{2.5} observed in other cities were in the range of 3.9 to $12 \mu\text{g m}^{-3}$, below the NAAQs. Despite the lower intensity of anthropogenic activities compared to Jakarta and Surabaya, the impact of forest fires in 2021 is generally weak, except in Pontianak.

In terms of diurnal variability, Padang, Palu, and Manado stand out from other cities. The two coastal cities located in Sumatra and Sulawesi Islands were affected by intense anthropogenic activities during certain hours. The high diurnal variation of PM_{2.5} hourly concentrations measured in Pontianak was mainly affected by the extreme differences between daily maxima concentrations and the average values especially during the forest fire events. Note that Jakarta and Surabaya are mainly anthropogenic emission-dominated cities, with 46% and 30% of the total emission load for PM_{2.5} in Jakarta in 2015 coming from road transportation and industrial combustion sectors, respectively (Lestari *et al.*, 2020). According to the Indonesia Statistic Bureau, every year, Jakarta recorded a significant increase in the number of fossil fuel vehicles from 16,491,682 in 2016 to 21,005,527 in 2021. On the other hand, Pangkalpinang has a low contribution from anthropological activities.

Maximum half-hourly PM_{2.5} concentrations recorded in the seven cities are tabulated in Table 2. Pontianak showed extreme concentration due to large forest fire events from the middle of February until early March 2021, where a 17,192-ha area in West Kalimantan Province was burnt (twice larger than the previous year). Meanwhile, Palu also registered a high concentration largely produced by combined emissions from vehicles, mining factories, and chimneys of steam power.

The diurnal pattern of the seven cities shows various characteristics, see Fig. 2. Padang, Manado, Palu, and Pangkalpinang present a similar diurnal pattern, with a morning and early night peak (bimodal pattern), which conforms with the worldwide pattern. Meanwhile, in Jakarta, Surabaya, and Pontianak, the diurnals have a unimodal pattern with one peak at night until morning. As addition, the COVID-19 pandemic occurred in 2021. At that time, the government imposed a Community Activities Restrictions Enforcement (lockdown) from 3 July until 6 September 2021

Table 2. Summary of PM_{2.5} of each location data from ground observations in 2021.

Cities	Total Area (km ²)	Total Population	PM _{2.5}		
			Mean $\pm$ SD	Diurnal variability	Max Half-Hourly
Jakarta	661.5	10,609,681	33.9 ± 24.0	18%	$190 \mu\text{g m}^{-3}$
Surabaya	326.8	2,880,284	27.1 ± 21.3	28%	$168 \mu\text{g m}^{-3}$
Padang	695	913,448	10.9 ± 9.8	41%	$97 \mu\text{g m}^{-3}$
Manado	157.3	453,182	11.9 ± 12.9	42%	$127 \mu\text{g m}^{-3}$
Palu	395.1	373,218	6.6 ± 6.2	21%	$175 \mu\text{g m}^{-3}$
Pontianak	118.3	302,735	23.4 ± 49.8	32%	$700 \mu\text{g m}^{-3}$
Pangkalpinang	89.4	221,988	3.9 ± 2.5	21%	$52 \mu\text{g m}^{-3}$

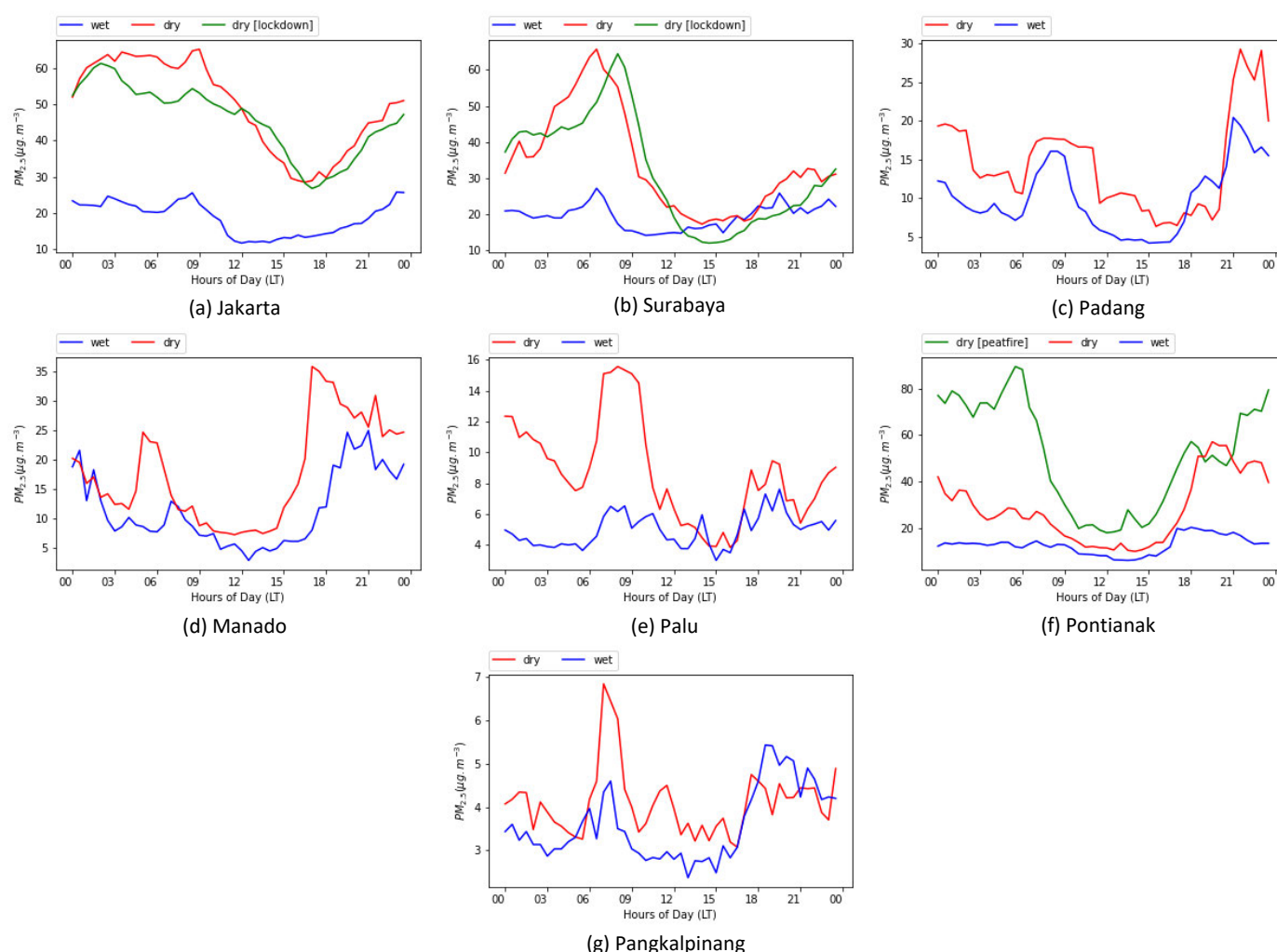


Fig. 2. The diurnal variation of $PM_{2.5}$ concentration in (a) Jakarta, (b) Surabaya, (c) Padang, (d) Manado, (e) Palu, (f) Pontianak, and (g) Pangkalpinang, during the dry and wet season.

in Java Island, including Jakarta and Surabaya. Therefore, the diurnal pattern of both cities for the dry season are separated into only lockdown data (green line) and without lockdown data (red line), see Figs. 2(a–b). The one peak in Jakarta and Surabaya indicates much greater human activity producing emissions during the night, leading to high late night-time concentrations, which is a common characteristic of metropolitan areas (Chen *et al.*, 2020c). Moreover, the locations of AQMS in Jakarta and Surabaya are in high-traffic volume areas (Ramdhan *et al.*, 2019). The peak time in Surabaya during the morning rush hours of the lockdown time was shifted to around one and a half hours later. While in Jakarta during the lockdown, the activities in the morning rush hours were significantly reduced, thus decreasing $PM_{2.5}$ daily concentration.

On the other hand, in Pontianak, with a smaller population, one peak in the dry season due to biomass burning occurred at around 06:00 in the morning, shown by the green line in Fig. 2(f), while during the normal day, the one peak took place at early evening depicted by the red line. During peatland fire, the daytime concentration of $PM_{2.5}$ is associated with the diurnal variation, whereas night-time enhancement was because of fire aerosols. Moreover, several hotspots were observed overnight despite the peatland fire occurring mainly during the daytime. In general, the lowest concentrations occurred in the afternoon (15:00–18:00 in Jakarta and 12:00–16:00 in other cities). The maximum concentrations took place in the morning (06:00–09:00) and the night (mainly starting at around 17:00). The morning peaks in Pontianak existed but were insignificant because of the AQMS location in urban green space and far from heavy traffic roads.



Table 3. Correlation between PM_{2.5} and PBLHs derived from the *Ri* and *PT* methods. The numbers in bold show statistically significant correlations at a 95% confidence level.

Period	Method	Jakarta	Surabaya	Padang	Manado	Palu	Pontianak	Pangkalpinang
Wet	<i>Ri</i>	−0.77	−0.79	0.67	0.51	−0.14	−0.27	−0.5
	<i>PT</i>	−0.78	−0.76	0.61	0.26	0.5	−0.14	−0.05
Dry	<i>Ri</i>	−0.12	−0.18	−0.94	0.37	−0.25	−0.49	−0.01
	<i>PT</i>	−0.36	−0.28	−0.31	−0.45	0.28	−0.77	−0.24

The monsoonal rainfall patterns with one peak during DJF contribute to different city characteristics. Jakarta and Surabaya (Figs. 2(a–b)) indicated a clear difference in PM_{2.5} between wet and dry seasons. Jakarta showed a high seasonal variation during the day and night, indicating high meteorological influence, while in Surabaya, a similar result was evident only in the morning. Meanwhile, Manado shows less variation, indicating a much lower influence (Fig. 2(d)).

An equatorial rainfall pattern with two rainfall peaks in Padang, Pontianak, and Pangkalpinang presented a higher PM_{2.5} concentration at night until early morning during the dry season than in the wet season (Figs. 2(c), 2(f), and 2(g)). However, the low rainfall variation throughout the year also contributes to a small difference in PM_{2.5} concentration between the dry and wet seasons. The significant rise of the concentration occurred in Pontianak because the forest fire events. The low anthropogenic activities and wind speed in Pangkalpinang led to low variation during all seasons, with the meteorological influence only obvious during the night. Meanwhile, Palu, which has an anti-monsoonal (local) pattern, showed little daily variation between both seasons except during morning rush hour, which showed a significant difference (Fig. 2(e)).

Ri and *PT* methods were applied for PBLH calculation to investigate its relationship with PM_{2.5}. Their correlation is computed using Pearson correlation. The agreement of both methods indicates the apparent correlation between PM_{2.5} and PBLH. The results tabulated in Table 3 show that each city has a different period of agreed correlation. A significant agreement on both methods occurred in Jakarta and Surabaya, where PM_{2.5} correlates negatively with PBLH during the wet season. Meanwhile, Padang and Pontianak show a strong anti-correlation in the dry season, indicating a significant descent of PBLH during nighttime, thus increasing the PM_{2.5} concentration. The other cities record insignificant correlations during both seasons.

3.2 Daily Variation and Combined Factors Explaining PM_{2.5} Variability

In this study, the effect of other meteorological factors on the daily and monthly PM_{2.5} pattern is further investigated (Table 4). The natural elements of meteorological parameters in an area can impose various effects on the amount of PM_{2.5}. The meteorological factors considered in this study are average temperature (T_{avg}), average relative humidity (RH), wind speed (WS), rainfall rate (RR), and sun duration (SS). The increased temperature in the lower height level will enhance the PM_{2.5} particle formation and influence the PBLH condition (Meng and Sun, 2021; Li et al., 2021; Hien et al., 2019). Humidity conditions influenced the hygroscopic chemical properties of PM_{2.5} (Wang et al., 2019; Wang et al., 2014).

Table 4. Pearson correlation between daily (r_{dd}) and monthly (r_{mm}) of PM_{2.5} with meteorological parameters.

Location	T_{avg}		RH		WS		RR		SS	
	r_{dd}	r_{mm}	r_{dd}	r_{mm}	r_{dd}	r_{mm}	r_{dd}	r_{mm}	r_{dd}	r_{mm}
Jakarta	0.32	0.59	−0.24	−0.58	−0.36	−0.60	−0.29	−0.69	0.19	0.53
Surabaya	0.11	0.47	−0.22	−0.52	−0.35	−0.54	−0.25	−0.85	0.17	0.71
Padang	0.45	0.89	−0.29	−0.56	0.14	0.46	−0.27	−0.30	0.16	0.44
Manado	−0.002	0.25	0.07	−0.16	−0.12	−0.04	−0.12	−0.19	0.15	0.22
Palu	0.20	0.24	−0.35	−0.73	0.10	0.50	−0.35	−0.77	0.07	−0.12
Pontianak	0.15	−0.08	−0.35	−0.60	−0.07	0.11	−0.24	−0.66	0.12	0.46
Pangkalpinang	0.12	0.39	−0.12	−0.27	−0.14	−0.07	−0.09	−0.02	0.20	0.49



Pearson correlation of daily and monthly mean PM_{2.5} concentration with each of the meteorological factors are presented in Table 4. The correlation is significant in monthly data and insignificant in daily data. Among the equatorial rainfall patterns, the PM_{2.5} concentration in Padang is highly influenced by temperature and slightly affected by rainfall rate; in the daily peak time, the concentration of the wet season is higher than in the dry season. Meanwhile, in Pangkalpinang, no meteorological factor is dominant because of the low concentration and its variation throughout the year. At the same time, the sunshine duration registers a moderate effect, resulting in low variability in all seasons. In Pontianak, rainfall rate has a significant influence in reducing the concentrations, except during early March when emissions from biomass-burning dominate. A similar characteristic was exhibited in Palu, which has an anti-monsoonal (local) pattern where, during the wet season, the rainfall decreased the PM_{2.5} concentration significantly. In the meantime, among the cities with a monsoonal rainfall pattern, only Manado is not significantly influenced by rainfall, thus, there is low variability. The highest rainfall intensity correlation occurred in Surabaya because of the significant difference in rainfall intensity between dry and wet seasons compared to other cities.

Pearson correlation analysis cannot find a significant daily correlation between a single meteorological factor and the concentration of PM_{2.5}. A study in China by Wang *et al.* (2020) revealed that the correlations between PM_{2.5} and individual meteorological factors are found at broader time scales. It suggested studying the joint effects of meteorological factors assumed to be the reason for their time scale-dependent correlations. A previous study by Li *et al.* (2021) exemplifies that using a combination of meteorological parameters could give some indication of what meteorological factors could control the PM_{2.5} concentration. Therefore, this study investigates correlation analysis of combination meteorological factors using MWC. Preprocessing in MWC was first conducted on the dataset to avoid misinterpretation in the wavelet. Next, the Morlet wavelet was chosen as the mother wavelet since the oscillation of the signal object (PM_{2.5} and the meteorological factors) is suitable with the complex sinusoidal function feature of the Morlet properties (Chen *et al.*, 2020c).

Three experiments were implemented in this study using MWC at each location. We used two, three, and four combinations of meteorological factors applied to the MWC. The analysis of AWC and PASC will provide a sound judgment on the relationship. The higher the value, the stronger the influence. Table 5 shows the MWC result between PM_{2.5} and two combinations of meteorological factors. The combination is represented by A1 until A10. Each location has a different coherence value concerning the AWC and PASC. Only Padang (A4), Pangkalpinang (A5), and Pontianak (A9) have the maximum value of both AWC and PASC in different combinations.

Further investigation is conducted by using three combinations of meteorological factors, with the result tabulated in Table 6. The combinations are represented by B1 until B10. Similar condition to the previous result, only a few shows the maximum value of AWC and PASC, almost similar result but with better PASC compared to (Li *et al.*, 2021). Observing the maximum and second maximum of AWC, we notice that B5 has the most occurrences (4) at Pontianak, Palu, Manado, and Surabaya. This condition was also followed by the most occurrence of PASC value for the maximum and the second maximum value.

Table 5. The AWC and PASC between PM_{2.5} and two combinations of meteorological factors.

Location	Combination of Meteorological Factors									
	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10
	AWC PASC	AWC PASC	AWC PASC	AWC PASC	AWC PASC	AWC PASC	AWC PASC	AWC PASC	AWC PASC	AWC PASC
Padang	0.75 12.37	0.73 16.37	0.72 13.86	0.76 14.59	0.67 8.71	0.73 15.33	0.74 15.93	0.65 7.50	0.72 13.78	0.70 8.73
Pangkalpinang	0.63 4.90	0.67 6.93	0.68 9.92	0.63 2.84	0.69 10.30	0.65 4.77	0.64 5.21	0.67 8.69	0.64 5.11	0.65 6.59
Pontianak	0.73 13.53	0.65 8.36	0.71 10.68	0.69 10.74	0.63 5.51	0.71 13.72	0.70 14.79	0.63 6.43	0.73 18.96	0.68 9.60
Palu	0.66 9.40	0.65 5.12	0.67 6.97	0.62 4.02	0.68 8.61	0.65 8.13	0.65 4.95	0.69 8.15	0.66 4.28	0.68 7.09
Manado	0.63 4.01	0.62 5.20	0.66 7.17	0.66 6.43	0.63 5.89	0.62 2.39	0.64 6.46	0.65 6.71	0.65 6.89	0.62 7.52
Jakarta	0.66 6.95	0.63 5.18	0.63 3.47	0.67 10.44	0.66 4.51	0.66 6.58	0.62 4.73	0.61 3.78	0.62 4.66	0.61 4.24
Surabaya	0.71 11.39	0.68 7.48	0.68 7.97	0.71 10.62	0.69 9.40	0.67 10.06	0.69 9.35	0.68 4.59	0.67 5.91	0.66 9.03

A1: T_{avg} -RH, A2: T_{avg} -WS, A3: WS-RH, A4: T_{avg} -SS, A5: WS-SS, A6: RH-SS, A7: T_{avg} -RR, A8: WS-RR, A9: RH-RR, A10: SS-RR.



Table 6. The AWC and PASC between PM_{2.5} and three combinations of meteorological factors.

Location	B1		B2		B3		B4		B5		B6		B7		B8		B9		B10	
	AWC	PASC	AWC	PASC	AWC	PASC	AWC	PASC	AWC	PASC	AWC	PASC	AWC	PASC	AWC	PASC	AWC	PASC	AWC	PASC
Padang	0.87	14.19	0.88	14.45	0.88	17.13	0.86	14.77	0.88	16.77	0.87	18.93	0.87	20.25	0.86	17.84	0.86	15.16	0.84	10.41
Pangkalpinang	0.83	9.95	0.80	4.27	0.84	10.49	0.84	11.68	0.81	4.23	0.83	10.83	0.80	4.25	0.83	9.81	0.81	4.83	0.83	8.58
Pontianak	0.85	12.42	0.86	19.80	0.82	10.09	0.83	14.23	0.87	22.22	0.83	11.67	0.83	13.78	0.85	17.41	0.84	17.77	0.81	11.68
Palu	0.83	10.04	0.84	11.26	0.82	6.41	0.83	9.36	0.85	16.36	0.82	9.49	0.82	9.44	0.83	9.17	0.83	10.51	0.84	9.44
Manado	0.80	7.25	0.82	6.61	0.81	7.40	0.82	8.53	0.82	9.06	0.80	6.54	0.83	7.99	0.82	6.80	0.80	7.98	0.81	7.97
Jakarta	0.82	6.13	0.84	12.29	0.82	10.73	0.82	6.93	0.82	7.73	0.78	5.24	0.81	8.68	0.81	4.87	0.80	4.55	0.80	5.49
Surabaya	0.84	10.77	0.85	11.16	0.85	13.69	0.84	10.89	0.86	10.87	0.84	8.78	0.85	11.04	0.83	7.98	0.84	11.90	0.84	8.81

B1 = $T_{avg}-RH-WS$, B2 = $T_{avg}-RH-SS$, B3 = $T_{avg}-WS-SS$, B4 = $RH-WS-SS$, B5 = $T_{avg}-RH-RR$, B6 = $T_{avg}-WS-RR$, B7 = $T_{avg}-SS-RR$, B8 = $RH-WS-RR$, B9 = $RH-SS-RR$, B10 = $WS-SS-RR$.

Table 7. The AWC and PASC between PM_{2.5} and four combinations of meteorological factors.

Location	C1		C2		C3		C4		C5	
	AWC	PASC	AWC	PASC	AWC	PASC	AWC	PASC	AWC	PASC
Padang	0.943	18.79	0.938	20.18	0.943	23.14	0.932	15.49	0.943	21.54
Pangkalpinang	0.918	11.65	0.918	11.73	0.899	7.04	0.918	9.6	0.92	10.76
Pontianak	0.92	18.04	0.924	20.03	0.928	20.75	0.917	17.36	0.912	13.76
Palu	0.927	17.45	0.925	16.21	0.931	17.81	0.92	11.63	0.911	8.61
Manado	0.913	9.7	0.908	12.06	0.915	12.82	0.904	10.1	0.911	8.02
Jakarta	0.92	14.42	0.91	7.15	0.913	10.3	0.903	6.93	0.905	9.25
Surabaya	0.926	13.61	0.929	17.9	0.934	16.95	0.929	16.5	0.929	22.44

C1: $T_{avg}-RH-WS-SS$; C2: $T_{avg}-RH-WS-RR$; C3: $T_{avg}-RH-SS-RR$; C4: $RH-WS-SS-RR$; C5: $T_{avg}-WS-SS-RR$.

The result using four combinations of meteorological factors is shown in Table 7. The maximum value of both AWC and PASC is in column C3 at Padang (0.943, 23.14%), Pontianak (0.928, 20.75%), Palu (0.931, 17.81%), Manado (0.915, 12.82%), and Surabaya (0.934, 16.95%). Jakarta has its maximum value of both AWC and PASC in column C1 (0.920, 14.42%), with the second maximum value spread in columns C3 (0.913, 10.3%). The utilization of four combinations can lead to the finding of the significant result of AWC and PASC in the C1, C3, and C5.

Fig. 3 shows the MWC power spectrum results from the significant combinations of C3 at the selected locations (Surabaya, Padang, and Palu). MWC consists of a response variable (PM_{2.5}) and a predictor variable (meteorological factors). The two MWC outputs are power spectrum and confidence level. The desired conditions are high coherence where the power spectrum value is near 1. The blobs (the thick black curve) are the areas with a coherence level of at least 95% between the response and the predictor variable at the corresponding period. The selection of Jakarta, Padang, and Palu in this analysis is to represent each rainfall pattern, which meteorological factors have the highest influence (see Table 4).

Observing the blob's location in Fig. 3 shows that the elongated blobs reside for 64–128 days (3–4 months) in a year. This condition showed that the PM_{2.5} was significantly correlated to the four combinations of meteorological factors of C3 at scales of around 3 to 4 months, at all times in Surabaya, Padang, and Palu showing the characteristics of coastal area in the tropical region. Correlation at scales of around two months (32–64 days) was also found in Palu throughout the year for C3 with PASC 17.81%, where local meteorological factors play a crucial role. Meanwhile, Padang shows a prominent correlation at scales of 2 to 4 months (64–128 days) throughout the year with PASC larger than 23%, which indicated a significant influence of the Barisan Mountain alongside the coastal area.

The overall result using various combinations of meteorological factors from two to four showed an increase of both AWC and PASC if many parameters are involved in the MWC. This condition indicated that the variation of PM_{2.5} concentration is well affected by multiple combinations of meteorological factors (Zhang et al., 2022b). Using the result from three and four combination experiments, the meteorological factor combination containing average temperature,

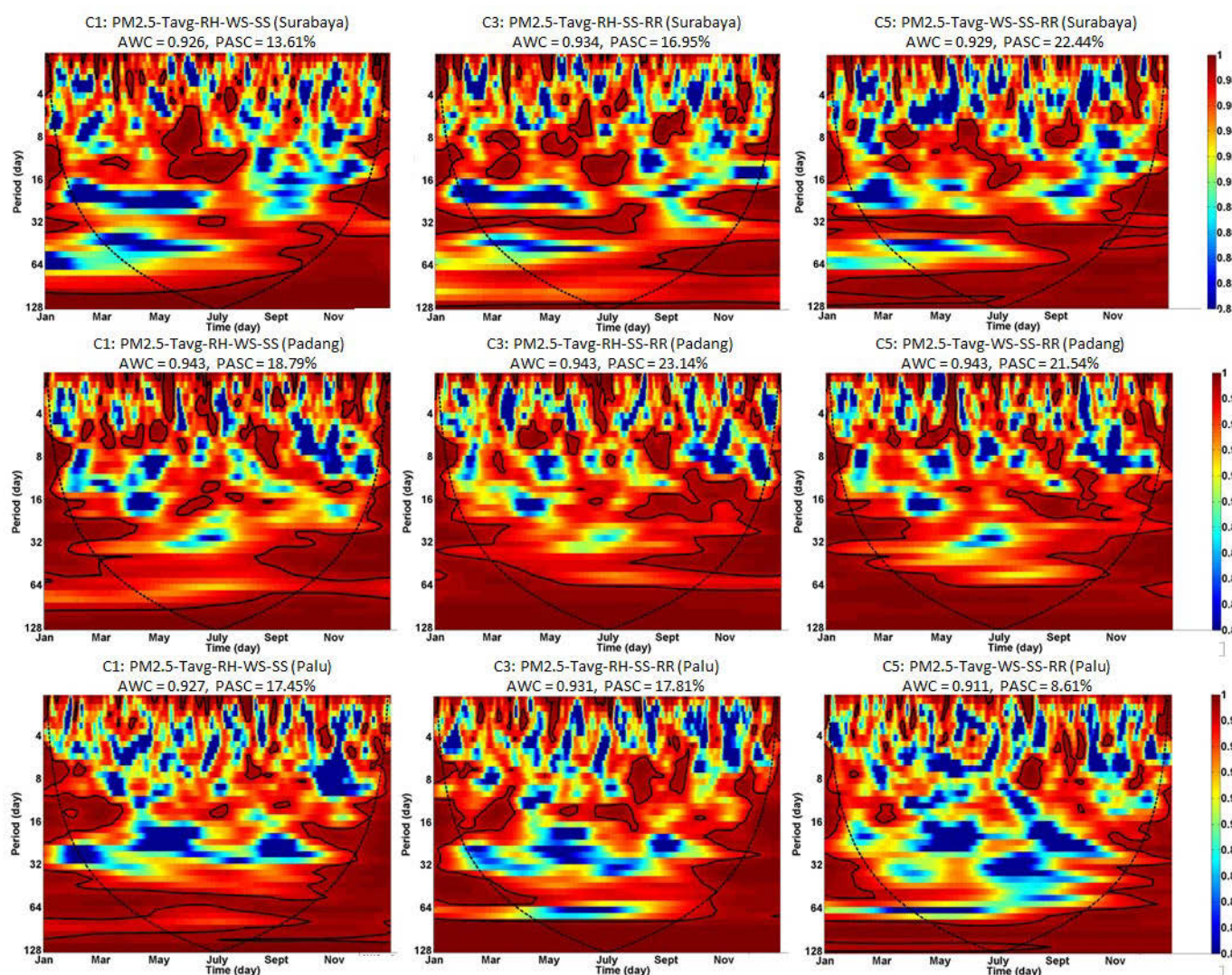


Fig. 3. MWC between PM_{2.5} and four combinations of meteorological factors (C1, C3, and C5). The X-axis is the time (day), and the y-axis is the period (day). The color represents the coherence level from 0.8–1, while the thin half circle is the cone of influence (COI). The blobs (thick black contour) have a high coherence value with a 95% confidence level representing the strong relation condition. The blob's location within the x and y axis contains the information on the when and period of the strong relation condition.

relative humidity, and rain rate strongly coincides with the PM_{2.5} concentration in almost all locations. Furthermore, the average temperature is more pronounced from two to four combination experiments.

4 DISCUSSIONS

The global diurnal pattern of PM_{2.5} is a bimodal pattern, as reported by Manning *et al.* (2018), while the unimodal pattern usually takes place in rural areas (Zhao *et al.*, 2009). On the contrary, Jakarta and Surabaya, known as metropolitan cities, showed unimodal patterns at late night and early morning, respectively. The unimodal patterns were caused by high emissions due to much greater human activity during the night. A similar diurnal pattern was also seen in Pontianak, with one peak at night. However, the effect of fire aerosols because of biomass burning has increased the average significantly, particularly in the early morning. The relationship between PBLH and PM_{2.5} is different in every city. They are commonly correlated negatively (Tursumbayeva *et al.*,



2022). Jakarta, Surabaya, Padang, and Pontianak show a strong anti-correlation for different seasons, while an unusual positive correlation occurred in Padang.

Previous studies have shown that different locations and lengths of datasets generate different results. *Li et al. (2021)* introduce the MWC result in Guiyang, China, using 5 years of observation data and two to four combination factors. They conclude that the combination of CO, average temperature, average wind speed, and relative humidity strongly relates to the PM_{2.5} variation. Different locations show different results. In Xi'an, China, the MWC revealed temperature, wind speed, and wind direction were the most substantial factors affecting PM_{2.5} (*Zhang et al., 2022b*). MWC within broader areas of urban agglomeration also showed different results. *Wu et al. (2023)* showed that a combination of precipitation, relative humidity, and temperature factors has the most robust relation to PM_{2.5} variation. Related to the present study in 7 different cities in Indonesia, the meteorological factors that strongly correlated with PM_{2.5} did not entirely correlate with the previous study. However, average temperature and wind speed are the common factors between the present and previous studies, although in different locations with different lengths of datasets. One thing in common between the present and previous studies is that MWC can demonstrate the multi-variation of meteorological factors about the PM_{2.5} variation.

Implementation of MWC with various meteorological combinations among five cities (Padang, Pontianak, Palu, Manado, and Surabaya) shows PM_{2.5} was significantly correlated to a combination of average temperature, relative humidity, and rain rate based on the result of B5 and C3 experiment at scales of around 3 to 4 months throughout the year in mostly all the observed locations showing the characteristic of coastal area in the tropical region.

Further investigation on PM_{2.5} concentration is related to haze pollution in metropolitan cities and peatland fires in Pontianak, which are described as follows.

4.1 Haze Case in Jakarta and Surabaya

Jakarta and Surabaya were often suffering from photochemical smog haze pollution. An extended two-year (2020–2021) daily dataset of PM_{2.5} was used to investigate the haze condition within these two locations. In this analysis, we utilize the real part of the wavelet coefficient. The real part of the wavelet coefficient is a tool to investigate the power spectrum distribution of wavelet coherence at different times and scales. This information is useful to analyze the fluctuation condition and the transition pattern of the daily PM_{2.5} concentration.

Fig. 4 shows the real part of the wavelet coefficient along with the time series of two years of PM_{2.5} concentration in two cities. There is a distinct scale of PM_{2.5} variation in scale 210–394 days (Surabaya) and 178–394 days (Jakarta). Within this scale period, the five-contour region of positive and negative represents the group of maximum or minimum PM_{2.5} concentration, which represents the haze condition. A positive value (yellow to red color) represents a higher haze condition which means that the haze condition is poor. Meanwhile, the negative value represents a low haze condition. There are two contours regions within two years of data. The positive regions were observed from May to September in both cities (indicated by A for 2020 and B for 2021) in accordance with the dry season. The transition period between high and low haze appears distinctively.

4.2 Peatland Fire in Pontianak

The green line in *Fig. 2(f)* shows a significant increase of diurnal concentration during the peatland fire event compared to normal conditions in Pontianak. According to the reports by local and electronic media in the period from mid-February to early March 2021, peatland fires spread into up to 40 hectares of the West Kalimantan area, creating 741 hot spots on 27 February 2021 (*Saputra, 2021*). The peatland fires started on 25 February 2021 in 14 areas, peaked one day later with 44 burnt areas in the north and south of Pontianak, and became extinct on 3 March 2021.

HYSPLIT with backward trajectory was performed on 28 February, 1 March, and 2 March 2021 to investigate the source of the extreme PM_{2.5} concentrations directed towards Pontianak city. The backward trajectory is executed with the mid-boundary layer height setting of 500 m, 1000 m, and 1500 m above sea level, and a reverse track for 6 hours using ensemble type at two-times domain, which is at 7 in the morning and 19 in the late afternoon. The 6-hour reverse track was chosen to see local influences at the monitoring point. The meteorological input data for the

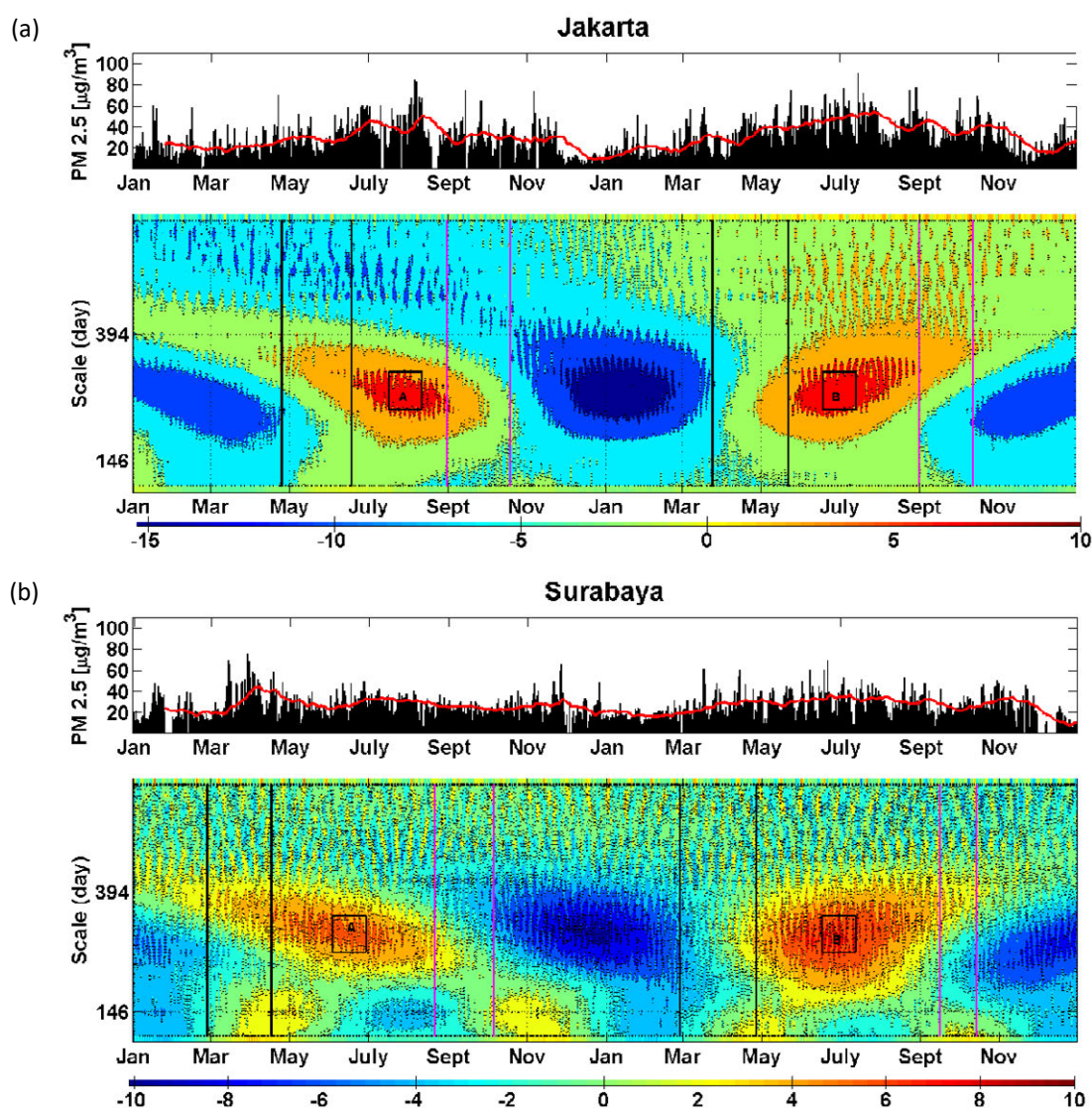


Fig. 4. The real part of wavelet coefficients $R\{w^{YX}(s, \tau)\}$ in two years of (a) Jakarta, (b) Surabaya. The y-axis is the concentration, with the red line representing the moving average of the $PM_{2.5}$ concentration. The lower panel is the real part of the wavelet. The horizontal axis is the time in the month unit, and the vertical axis is the scale in the day unit. Box A and B represent the high haze conditions in 2020 and 2021, respectively. The vertical line (black and purple) represents the transition period between low haze and high haze.

HYSPLIT is obtained from NCEP GDAS (Global Data Assimilation System). The calculated trajectories contain the various start points, the endpoint (location of the $PM_{2.5}$ observation), and the trajectory line. The results are visualized in Fig. 5.

On 28 February 2021 at 07:00 LT (Fig. 5(a)), the sources originating from the north, west, and south brought $PM_{2.5}$ to the city and raised the concentration to $254 \mu g m^{-3}$. At 19:00 LT (Fig. 5(b)), the source came from a similar direction, but the increase from the south inflicted $345 \mu g m^{-3}$ of cumulative concentration. One day later, at 07:00 LT (Fig. 5(c)) and 19:00 LT (Fig. 5(d)), the pollutants arrived from the west and north, producing $562 \mu g m^{-3}$ and $110 \mu g m^{-3}$, respectively. Meanwhile, on 2 March 2021, the source dominantly originated from the north and some part from the nearby south, both at 07:00 LT (Fig. 5(e)) and 19 LT (Fig. 5(f)), generating a cumulative concentration of $700 \mu g m^{-3}$ and $136 \mu g m^{-3}$, respectively. As a result, the maximum half-hourly $PM_{2.5}$ concentrations in Pontianak reached $700 \mu g m^{-3}$. Thus, the average concentrations during the dry period also rose while the diurnal peak increased, as shown by the green line in Fig. 2(f).

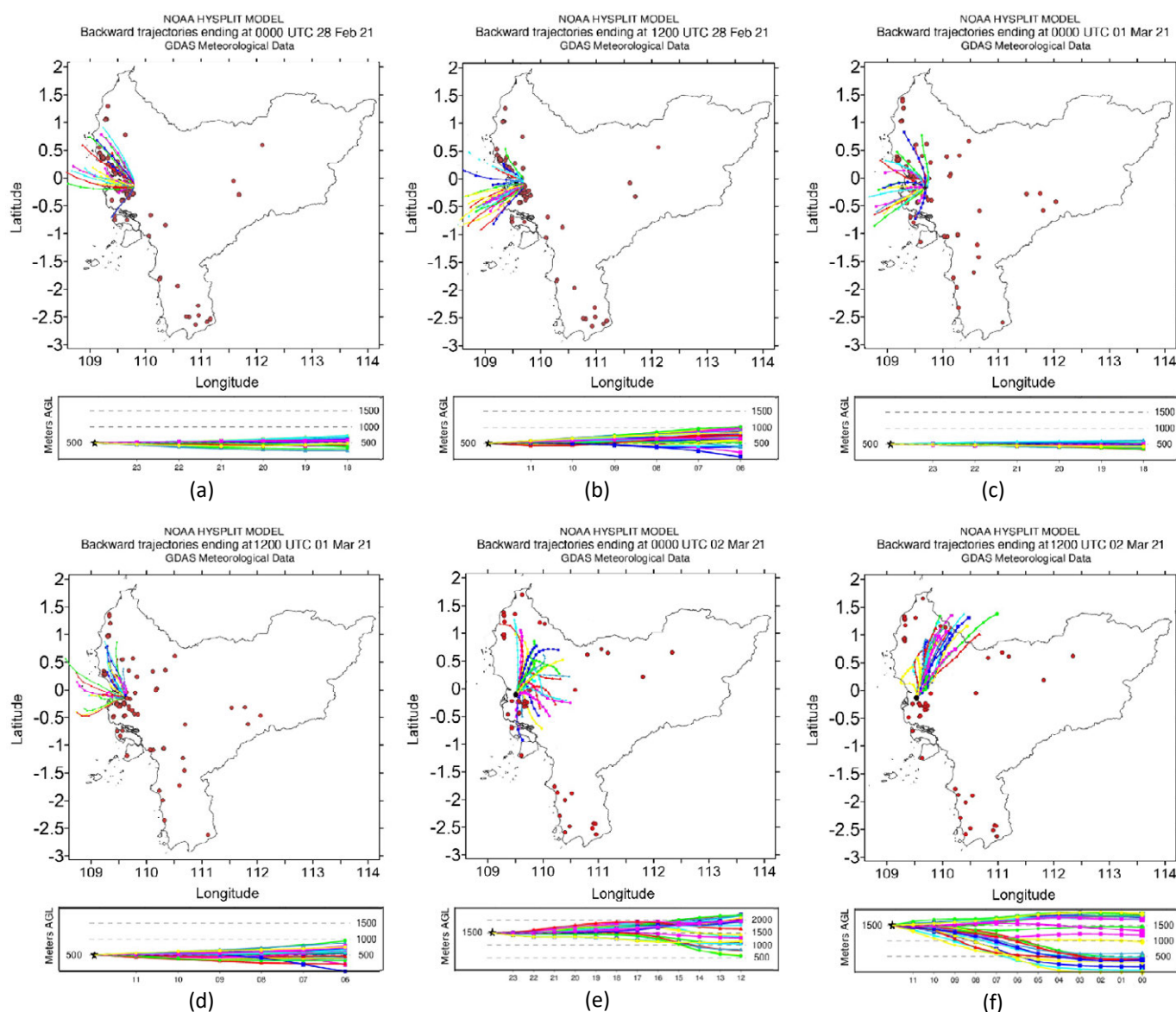


Fig. 5. The calculated trajectories of pollutants toward Pontianak projected using HYSPLIT on 28 February 2021 at (a) 07 LT, (b) 19 LT; on 1 March 2021 at (c) 07 LT, (d) 19 LT; on 2 March 2021 at (e) 07 LT, (f) 19 LT. The trajectories incorporated with movement in altitude along the time.

5 CONCLUSIONS

This study uses multiple datasets from several $\text{PM}_{2.5}$ monitoring, radiosonde, and meteorological stations to investigate the diurnal variation of $\text{PM}_{2.5}$ and its relationship with meteorological variables during 2021 in seven cities of Indonesia. The maximum half-hourly $\text{PM}_{2.5}$ in Pontianak is the highest among others due to a peatland fire event. Padang, Manado, Palu, and Pangkalpinang present a similar diurnal pattern to the worldwide pattern. Meanwhile, in Jakarta, Surabaya, and Pontianak, the diurnals have a unimodal pattern with one peak at night or morning. The cities with monsoonal rainfall patterns do not show similar characteristics of diurnal patterns, while the cities with equatorial rainfall patterns similarly present a higher $\text{PM}_{2.5}$ concentration at night until early morning during the dry season than the wet season. The anti-monsoonal (local) pattern city shows little daily variation between the dry and wet seasons except during morning rush hour which shows a significant difference.

The relationship between PBLH and $\text{PM}_{2.5}$ is different in every city. Jakarta, Surabaya, Padang,



and Pontianak show a strong anti-correlation for different seasons, while an unusual positive correlation occurred in Padang. The Pearson correlation of daily and monthly mean PM_{2.5} concentration with each meteorological factor is significant and insignificant in monthly data. Implementation of MWC with various meteorological combinations among five cities (Padang, Pontianak, Palu, Manado, and Surabaya) shows PM_{2.5} was significantly correlated to a combination of average temperature, relative humidity, and rain rate at scales of around 3 to 4 months throughout the year. Wavelet analysis also observes distinct scale periods representing higher haze concentrations in Jakarta (May–September) and Surabaya (April–September) on the scale of 178–394 days and 210–394 days, respectively. Meanwhile, an extreme concentration event occurred due to peatland fires in the West Kalimantan area, showing that sources from the north and south contributed to the extreme rise of the concentration in Pontianak by recording a maximum half-hourly concentration reaching 700 $\mu\text{g m}^{-3}$.

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